

Research Interests

Calibrated uncertainty for frozen foundation models under distribution shift, applied to climate and urban heat. My work asks whether published reliability guarantees survive the shift a deployed model actually meets, and so far the answer has been no. My current focus is Earth-observation foundation models, and whether robustness benchmarks that measure accuracy would detect a reliability failure at all.

Education

National University of Computer and Emerging Sciences (FAST-NUCES) 2020 – 2024
 Bachelor of Science in Computer Science. Degree awarded 28 June 2024. Islamabad, Pakistan
 – Gold Medallist and Rector’s List of Honour, seventh semester. Batch rank 57 of 285. Grade average across the final 30 credits: 3.60/4.00, including a 4.00 semester.

Preprints and Papers Under Submission

All single- or two-author, produced outside any institution on a laptop from public data, with no paid compute. Every result is reproducible from the linked repository.

A Frozen Foundation Model Is No Better Calibrated Than Six Spectral Indices: Auditing Urban Heat Maps Across Cities

A. Javed

Preprint, 2026. Under submission to Climate Change AI @ NeurIPS 2026 (Papers track). doi:10.5281/zenodo.21853486 | [code](#)

Across four cities on three continents, city-wide conformal coverage looks valid at 0.888 to 0.901 while per-neighbourhood coverage spans 0.736 to 0.978 and city-to-city transfer falls to 0.105. A frozen 1024-d Clay v1.5 embedding is no better calibrated under transfer than six hand-built spectral indices, 0.513 against 0.480. A prediction registered in advance was falsified and is reported rather than dropped.

Calibrated Station-Scale Temperature Downscaling from a Temperature-Agnostic Geospatial Foundation Model

A. Javed and B. Javed (FAST-NUCES)

Preprint, 2026.

doi:10.5281/zenodo.21849728 | [code](#)

Frozen, leakage-free Clay v1.5 embeddings cut RMSE by 38 to 68 percent over statistical downscalers across three climate regimes. Split-conformal intervals are near-nominal in-regime, then collapse from 90 to roughly 12 percent under Arizona to Pacific Northwest transfer. Group-conditional (Mondrian) conformal partially restores validity.

When Does a Frozen Time-Series Foundation Model’s Uncertainty Fail, and What Actually Fixes It?

A. Javed

Preprint, 2026.

doi:10.5281/zenodo.21848966 | [code](#)

Native intervals collapse under regime shift, not horizon. No static post-hoc conformal method repairs conditional coverage, while an oracle shows the failure is calibration rather than difficulty. Online conformal recovers extreme-drift coverage from 48 to 65 percent, and a leakage control shows the result is not contamination.

Research Experience

Independent Researcher

2025 – Present

Calibrated uncertainty under distribution shift

Remote

- Designed and ran every experiment above end to end: problem selection, literature gap verification, data acquisition from public sources, implementation, evaluation and writing.
- Built a model-agnostic, unit-tested conformal prediction library (split-conformal, CQR, Mondrian, adaptive conformal inference) reused across all three studies.
- Adopted a practice of registering the expected result before running the experiment, so that a falsified prediction is reportable evidence rather than a discarded run. One such prediction has been falsified and published as a finding.

AI/ML Engineer

Jan 2025 – Present

FortyGuard, urban heat and climate intelligence

Abu Dhabi, UAE (Remote)

- Designed a spatiotemporal temperature model covering eight or more U.S. cities at 10 m resolution, the applied setting from which the reliability questions in the work above originally arose.
- Built multi-source geospatial pipelines processing more than 10 million observations per day, and own model evaluation and drift monitoring in production, which is where the gap between reported accuracy and deployed reliability became concrete.

Machine Learning Engineer Intern

Jun 2023 – Sep 2023

Payactiv

Islamabad, Pakistan

- Built gradient-boosted risk scoring over more than 1 million transaction records, where calibrated abstention mattered more than raw accuracy.

Technical Skills

Geospatial: Sentinel-2, Landsat Collection 2, Microsoft Planetary Computer, rasterio, cloud-optimised GeoTIFF, local climate zones, coordinate reference systems

Uncertainty and evaluation: conformal prediction (split, CQR, Mondrian, adaptive), calibration assessment, coverage and interval-width analysis, distribution-shift evaluation, leakage-controlled experimental design

Machine learning: Python, PyTorch, scikit-learn, XGBoost; foundation-model representation probing, time-series forecasting, spatiotemporal modelling

Research engineering: reproducible pipelines, Docker, CI/CD, LaTeX, Git. Languages: English (IELTS Academic 7.5, February 2025), Urdu (native).